

Yupeng Lu

US Permanent Resident · No sponsorship required
yplmicro@gmail.com · linkedin.com/in/yupeng-lu-845a0b411
GitHub: github.com/BrickerP · Public work: github.com/cookiy-ai

AI agent & backend engineer — I take agents from demo to production: MCP tools, guardrails, evals, billing, real-time systems.

SUMMARY

AI agent and backend engineer who ships the measurements with the code. Agent platforms at Baidu on MeDo (agent-native infra access, release guardrails, Stripe billing-service extraction) and Cookiy (41 tools across 4 MCP servers, a 291-case eval harness, evidence-gated releases); a live real-time trading system on Alpaca that publishes its latency numbers with their method; and a deterministic 48-second WebGL film with CI performance and seam gates.

PUBLIC PROOF

- [public TypeScript CLI](#) / [npm CLI package](#) - AI interviews, synthetic users, surveys, recruitment, playback, and billing via REST/MCP; [MCP package](#) and [developer portal](#) with one-command skill/MCP setup and Redoc/OpenAPI docs.
- [public user-research skill](#) - AI-agent skill for end-to-end user research across Claude, Codex, Cursor, and OpenClaw-style clients.
- [technical sheet 01 - production real-time system](#) - order path, execution-safety invariants, and the 2026-10-08 scan-latency regression with its method box; [public research dashboard](#).
- Side projects: [fed-pulse](#) and [auction-tape](#) - stdlib-only Python CLIs over public macro data with offline tests and CI.

EXPERIENCE

Baidu - Software Engineer, MeDo

Jul 2026 – Present · Beijing

[MeDo](#) turns a chat prompt into a full-stack app that coding agents generate, deploy, and host. Overseas engineering owner for the revenue platform (Stripe billing, credits ledger, ads attribution, growth, SEO) and builder of the agent tooling the team uses to ship and operate it. 600+ merged code changes across 19 repos and 140+ design/RCA/verification docs in under 3 months.

- **Agent-native infra access.** Authored medo-infra-connect, a coding-agent skill + unified CLI (~5.5K LOC) adopted in the team skill repo: governed agent access to AWS, MySQL, ClickHouse, OpenSearch, Metabase, and Google/Meta/Bing/GA4 APIs across US and China environments; read-only by default, a verified prod-write path, per-user OAuth, no secrets in the repo.
- **Agent workflow guardrails.** Built release-proof skills (code review → deployed bundle → design doc, 14 incident-derived gates) and a policy hook (59-case suite) enforcing pipeline-only builds in Claude Code, Codex, Cursor, and OpenCode; wired codegraph MCP (48 repos), Opus-to-Codex delegation, persistent memory, and an MCP schema-fix proxy for the company LLM gateway.
- **LLM operations bots.** Took over and hardened the Discord → LLM-enriched report → intake desk → issue-tracker pipeline (durable capture + outbox, bounded LLM budgets, AI prefill, guarded rollback; cleared an 8-day ticket stall; merged 3 services, 788 tests); built an IM data-assistant bot whose skills query a read-only Metabase proxy (no free-form SQL).
- **Agent sandbox abuse response.** Contained a cluster-wide outage in which bot accounts prompted the coding agent to install remote shells and run crypto-miners in Kata-VM sandboxes: quantified 42× CPU overcommit, re-weighted node system cgroups, killed miners, and blocked pools by IP; worst-node CPU 96% → 48%, load 269 → 52, ~40 cores reclaimed, zero restarts.
- **Billing service extraction.** Led the move of the overseas credits ledger and Stripe stack (27 webhook event types, ~29K lines) out of the monolith into a new billing service on its own RDS; wrote the plan, runbook, and 28-item SQL acceptance matrix; caught ~10 rehearsal blockers (e.g., DMS showing 100% with 6 tables suspended); cutover loaded 40/40 tables with 0 errors.
- **Subscriptions and pricing.** Shipped Pro Max and annual plans on a two-phase quote → commit plan-change API over Stripe Subscription Schedules (20 countries, 10 currencies), Billing Portal cancellation with retention offers, past-due self-serve recovery, promo codes on every checkout path, and a first-purchase offer A/B (+88% lift in 48-hour first-payment rate).
- **Billing correctness.** Root-caused Stripe lifecycle defects (canceled subscriptions revived by late invoices, ~240 locked accounts; unsynced renewals; upgrade declines misread as downgrades; swallowed fulfillment rollbacks); fixed them with terminal-state gates, Stripe-status checks, and retry → dead-letter state machines; moved compensation to self-expiring \$0 subscriptions.
- **Ads attribution and consent.** Found one-time top-ups had never been uploaded to Google Ads (a Spring afterCommit write silently joining a committed transaction) and fixed it; built Meta Pixel + CAPI, Bing UET server-side conversions via a transactional outbox, and a first-party CMP (EEA/UK/CH opt-in, CA opt-out + GPC) gating 5 channels server-side; converged on an append-only one-winner-per-user model with a registration-time gate (22% of winners had been polluted).
- **Credits economy and abuse.** Cut gifted credits 35% (~58% per new user) without hurting retention or paid conversion; moved invite rewards to publish/payment milestones (spend down two-thirds); caught 437 inviters farming 20.6K fake accounts and shipped risk scoring with gifted-only clawback; fixed phantom debt entries and an expiry job skipping half its rows.
- **Data and SEO.** Built K8s CronJob collectors (Google Ads, Bing, GA4, Search Console, logs) → ClickHouse → Metabase dashboards; diagnosed an indexing collapse (crawl budget eaten by ~18K UGC app pages); shipped a Next.js 16 blog replacing Ghost (/blog payload -92%, crawlable posts 12 → 59); moved a 1.9K-URL pSEO site from a hand-run EC2 box onto EKS CI/CD.
- **Enterprise and Azure OpenAI.** Built enterprise Seat Pool + partner redemption codes across 6 repos (launched with a K-12 education partner) and Azure OpenAI resale hosting with per-customer resource isolation.

Cookiy AI - AI Agent Engineer

Aug 2025 – Jul 2026 · Silicon Valley HQ / Beijing Engineering Team

Voice-AI platform for user-research interviews. Built agent-native research infrastructure across MCP tools, public REST APIs, OAuth/billing, report generation, Report Voice Platform, interview/playback flows, and E2E/release gates; 1,040 merged PRs.

- **Contract-first MCP/tool platform.** Owned 41 tools across 4 MCP servers and 26 Zod/OpenAPI schemas (study, interview, quant, billing, recruit, guide, playback, report); hardened runtime validation, status semantics, and CLI payload compatibility.
- **CLI/skill distribution.** Published public skill and TypeScript CLI surfaces; migrated shell workflows to an npm CLI with MCP-first/REST-fallback behavior, token/login flows, Cursor/Claude deep links, and Homebrew installer distribution.
- **Tool-using chat agents.** Re-architected report/study agents around scoped registries, load-before-edit artifact discipline, evidence bundles, direct-save tools, and locked-fact guards; removed 7 duplicate tools and slimmed prompts by ~2.5k LOC.
- **Evidence-bound report synthesis.** Replaced single-shot prompting with objective-first multi-pass synthesis (cohort batches, objective-independence and emergent-discovery passes), chunked HTML fidelity/readiness gates, screener/chat corpus QA, and report-driven auto clips (MediaConvert + GCP Transcoder fallback).
- **LLM reliability and streaming.** Hardened OpenAI/Gemini paths (native fetch, uncompressed SDK responses, preserved auth headers, runtime pinning, Cloudflare/401 handling) and SSE/WebSocket/MCP streamable-HTTP long tool work (heartbeats, abort isolation, hard timeouts, run-ID propagation).
- **Eval/release infrastructure.** Built a dual-surface E2E harness (291 cases: 168 SaaS + 123 CLI/MCP), 7 L0-L6 gate profiles, preview SaaS/voice/share-link oracles, runtime evidence packets, trace correlation, p95 budgets, and a baseline ledger.
- **Realtime voice and Report Voice Platform.** Worked across LiveKit rooms, Deepgram STT, TTS providers, Gemini/OpenAI LLM paths, and Tavus/SpatialReal avatars (~200 ms barge-in); shipped a report-voice runtime with streaming TTS, speculative follow-up turn-taking, VAD barge-in, and voice-role training.
- **Agent-compatible auth and billing.** Hardened compact CLI tokens, OAuth/Bearer semantics, wallet-ledger idempotency, StripeEventLog webhook audit, pay-before-reveal report billing, and minimum-disclosure tool responses.

Quant Trading Systems Venture - Co-Engineer

Apr 2026 – Present · Concurrent

Private live-trading system on AWS: intraday scanner → factor filter → Alpaca bracket orders → SQLite execution ledger → S3 Parquet warehouse → [public research dashboard](#); 250+ commits since Jul 2026.

- **Execution safety on Alpaca.** Shared execution ledger with fill-dedupe keys, broker + DB startup reconcile, bracket/OCO fixes, and a flag-gated migration of live routing onto Alpaca. Retired path: IBKR heartbeat/reconnect and the IB Gateway route, removed after cutover.
- **Scan latency and SQLite write contention.** Event-based scanner waits, warm-cache fast paths, SQLite parse optimization, and lazy-load startup; on 2026-10-08 traced a scan-batch p95 regression (~2.4 s → ~9.2 s) to a 5-minute backfill cron holding SQLite's write lock and removed it on 2026-10-09.
- **Immutable public dashboard and ops.** Dashboard renders as a pure function of content-addressed input snapshots with Lambda publication (CAS supersede, skip-unchanged), redacted live logs, and CloudFront/S3 OAC hardening; rebuilt infra in a new AWS account with a verification checklist, SSM Parameter Store secrets, SSM-only deploys, and an after-close watchdog.

SMU - Software Engineer / 0-to-1 Product Owner

Feb 2024 – Jan 2026 · Los Angeles, CA

Data-platform role that expanded into a founder-sponsored 0-to-1 product: after shipping core commerce/ERP infrastructure, independently built and operated GetDateLove as a production consumer product.

- Built Python AsyncIO pipelines connecting Amazon Seller Central + Shopify with ERP, processing 500k+ daily orders and cutting sync latency from 15 min to 30 s; shipped Redis/Kafka inventory sync, FastAPI/PostgreSQL tariff services, Elasticsearch HS Code lookup, Spark migration jobs, and Docker/Jenkins CI/CD on AWS.
- Built GetDateLove end to end: FastAPI, redis.asyncio, PostgreSQL, WebSocket messaging, S3 voice notes, PayPal checkout, AWS/Nginx/CloudFront, CI/Playwright release gates, and GDPR/privacy controls.
- Reduced operational and frontend entropy by replacing blocking Redis patterns, consolidating 11 Redis clients, batching SQL, moving message storage to redis.asyncio, and extracting stable frontend state owners.

Huawei Technologies - Software Engineer, SPU Development

Apr 2023 – Feb 2024 · Beijing

- Built C/C++ SPU/router features for NE40E traffic statistics and NetStream monitoring; fixed 50+ packet-processing, flow-log, URL parsing, HTTP header, and SmartNet resource issues.

SKILLS

Agents and LLMs: MCP servers/tools, agent skills and CLIs, Claude Code / Codex / Cursor / OpenCode harnesses, Zod/OpenAPI tool contracts, evals and release gates, OpenAI, Gemini, Bedrock, Azure OpenAI, LiveKit voice.

Backend and data: TypeScript/Node (NestJS, Express), Python (FastAPI, AsyncIO), Java 21 / Spring Boot (JPA, MyBatis), C/C++; MySQL, PostgreSQL, Redis, ClickHouse, Kafka, Metabase.

Cloud and payments: AWS (EKS, RDS, DMS, Lambda, CloudFront, S3), Kubernetes, Stripe Billing, Google Ads/Meta CAPI/Bing UET.

EDUCATION

Boston University - M.S. Mathematical Finance & FinTech

GPA 3.78

Computational Methods, Stochastic Calculus, Algorithmic Trading

The Ohio State University - B.A. Physics

GPA 3.84 · Dean's List